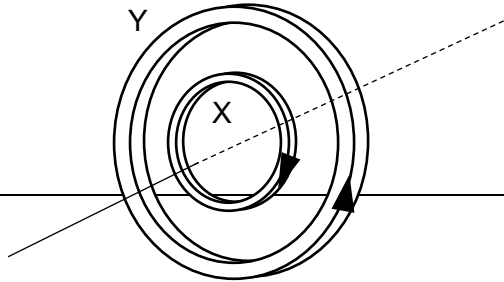


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Two flat circular coils, X and Y, each with 50 turns, are arranged as shown in the diagram.



[Turn over

X has radius 0.050 m and carries a current of 2.0 A. Y has radius 0.10 m and carries a current 4.0 A in the opposite direction to X. What is the magnitude of the total magnetic flux density at the centre of the coils?							
A	zero	B	$2000\mu_0$	C	$3000\mu_0$	D	$4000\mu_0$

Ans: A

$$B = \frac{\mu_0 NI}{r}$$

$$B_X = \frac{\mu_0 50 \times 2.0}{0.050} = 2000\mu_0$$

$$B_Y = \frac{\mu_0 50 \times 4.0}{0.10} = 2000\mu_0$$

$$\text{Total } B = B_X - B_Y = 0$$